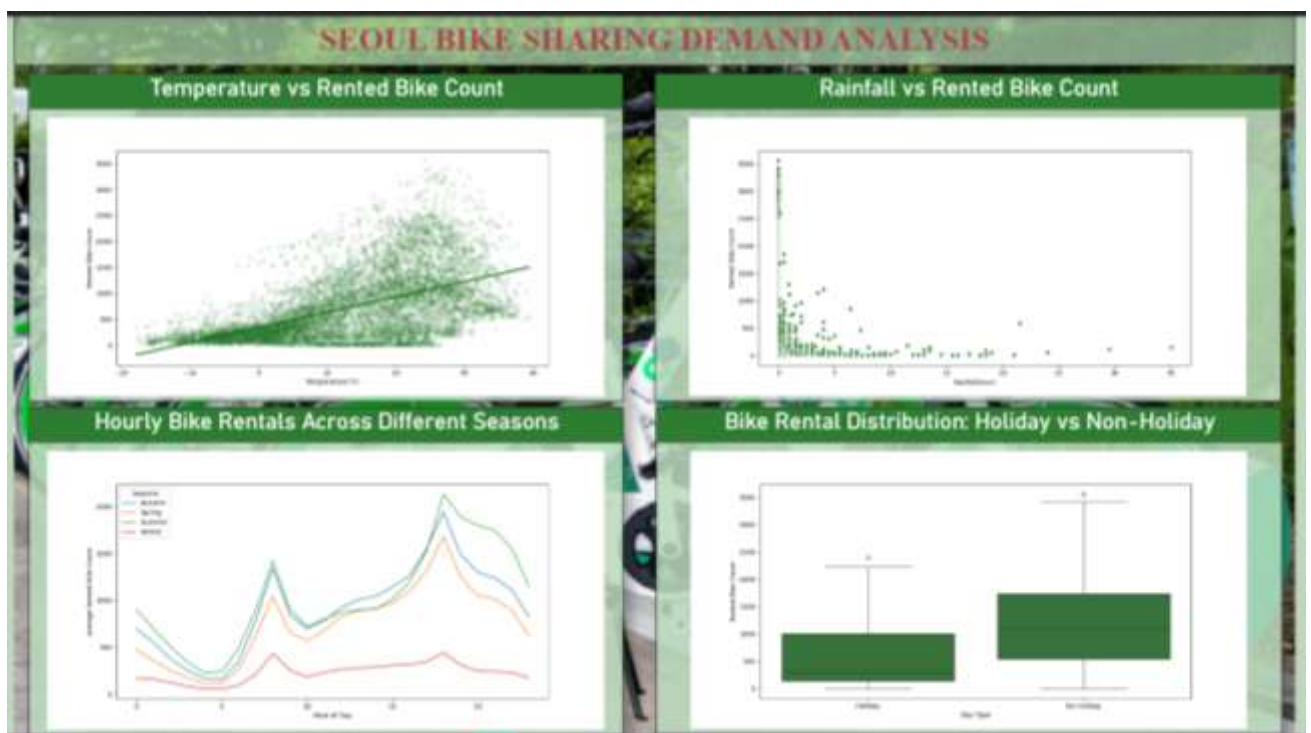
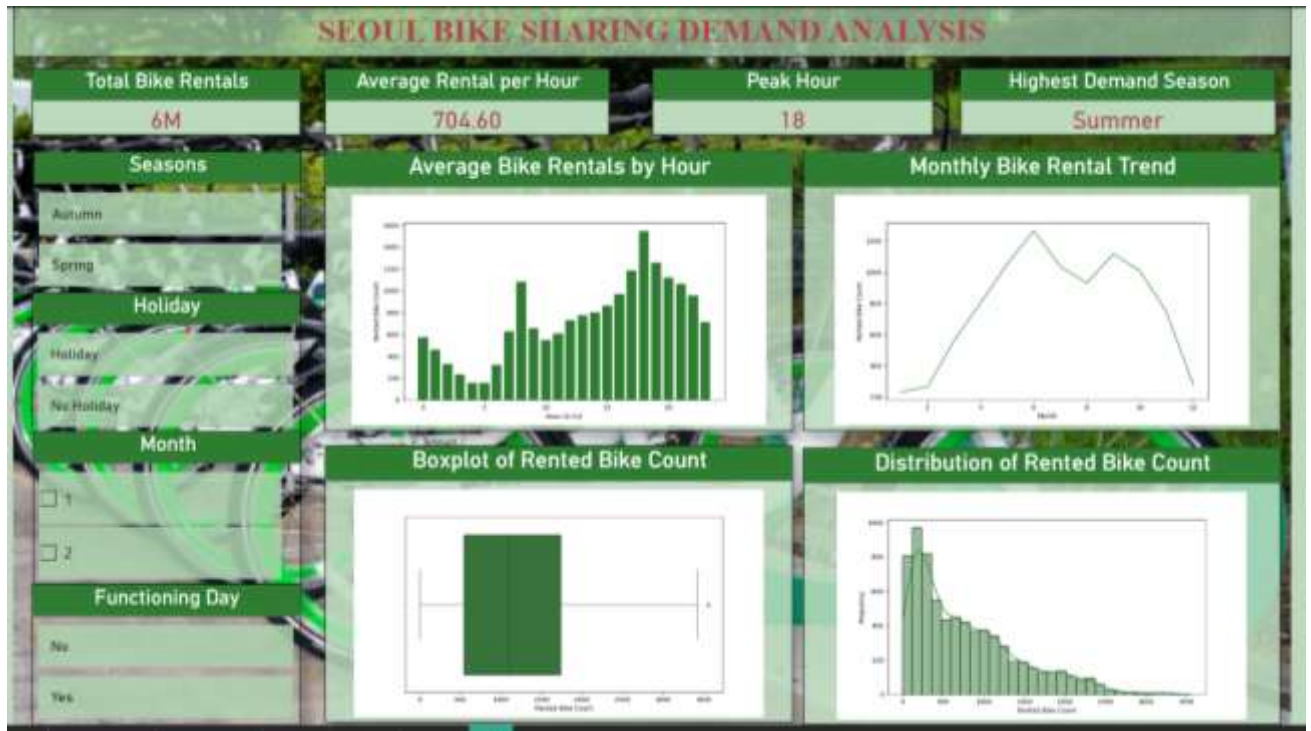


# SEOUL BIKE SHARING DEMAND ANALYSIS

## Power BI Dashboard





The Power BI dashboard gives a complete view of bike rental demand patterns in Seoul by combining time-based trends, weather conditions, seasonal effects and operational factors.

- The first dashboard shows the overall demand trends by hour, month, season and holiday.
- The second and the third dashboard focus on weather impact, seasonal variation and system availability on bike rental.

### Findings Summary:

- Bike rental demand in Seoul is not uniform and changes significantly across seasons, hours and weather conditions.
- Rentals demand peak during the mornings and evenings, showing a strong commuter usage.
- Warm and dry weather conditions increases bike usage, whereas rainfall and snowfall reduce demand.
- The usage of bikes peaks during summer and reduces during winter indicating seasonal patterns.
- Bike rentals are mainly driven by functioning days and non-holidays, highlighting the commuting behaviour rather than for leisure reasons.

### Key Findings:

- Bike rentals peak during office commute hours, indicating that bikes are mainly used for daily transportation.
- Weather is the most influencing external factor, with temperature having a positive correlation with bike rentals, while rainfall and snowfall have a negative impact.

- Seasonal demand is highest in summer, while it drastically reduces during winter due to unfavourable weather conditions.
- Non-holidays have higher demand, proving bikes are used more for routine travel.
- Rentals drop to almost zero on non-functioning days, indicating that system availability directly influences the bike demand.

### **Business Recommendation:**

#### **1. Resource Optimization**

Bike availability shall be increased during peak commute hours and summer months, and reduce supply during low-demand period.

#### **2. Weather-Based Planning**

The bike distribution and maintenance schedules can be adjusted based on weather forecasts, especially during heavy rain or snow.

#### **3. Seasonal Maintenance**

The major maintenance and repairs shall be planned during low-demand period like winter months.

#### **4. Commuter-focused Policies**

Weekly commuter passes or subscriptions can be introduced to target primary user base and improve customer retention.

#### **5. System Reliability**

Ensure maximum system uptime on working days, as even short periods of downtime can lead to a greater loss in demand.

### **Conclusion:**

The dashboard clearly shows how bike rental demand works in Seoul.

Bike usage is highest during **morning and evening office hours** and increases in **warm, dry weather**.

Working days have much higher demand than holidays, showing that rentals are mainly driven by daily commuting.

Seasonal trends indicate **peak usage in summer** and **lowest demand in winter** due to weather conditions.

Overall, these insights help service providers **optimize bike availability, plan maintenance**, and support better **urban transportation planning**.